



⚠ INCOMPLETE — SUPPORT-TIER

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

SDDC Manager Backup — System Config Fails on primaryDatastoreName

Root-cause and attempt record. The metadata export is blocked by a cluster inventory-relationship desync (KB 429861). NOT resolved — documented for handoff.

SDDC MANAGER

FILE-BASED BACKUP

PRIMARYDATASTORENAME

INVENTORY DESYNC

KB 429861

INCOMPLETE

v1.0

VMware Cloud Foundation 9

SDDC Manager Backup — System Configuration Fails on `primaryDatastoreName` [INCOMPLETE]

Prepared by: Virtual Control LLC **Date:** June 9, 2026 **Document Version:** 1.0 **Classification:** Internal **Status:** ●
INCOMPLETE — root-caused, not resolved. Support-tier (KB 429861).

This is not a fix — it is a diagnosis-and-attempt record. The SDDC Manager file-based backup's *System Configuration* step fails on a Python `KeyError: 'primaryDatastoreName'`. The platform database is **already correct**; the failure is a deeper **inventory-relationship desync** that the inventory API exposes, and Broadcom's published resolution (KB 429861) is **"contact Support."** It is documented here so the trail is preserved for a Support case or a later inventory reconciliation — and so the verified facts (values, IDs, log paths) don't have to be rediscovered.

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1. Symptom & Scope

- **UI:** *Backup SDDC Manager System Configuration → Failed.* Message: *"Backup operation failed for SDDC Manager System Configuration."* Cause: *"Exporting SDDC Manager metadata operation failed."* Remediation shown: *"Retry triggering backup again"* (retry does **not** help — same failure every run).
- **The failure is isolated to one sub-task.** The backup workflow has 8 steps; the database and configuration backups **succeed**, only the metadata step fails:

#	Sub-task	Result
1	QuiesceSystem	✓ Successful
2	BackupSDDCManagerDatabase	✓ Successful
3	BackupSDDCManagerConfiguration	✓ Successful
4	BackupSDDCManagerSystemConfiguration	● Failed — BACKUP_SDDC_MGR_SYSTEM_CONFIG_FAILED
5–7	Package / Save / Delete-old	⏸ Pending (never run)
8	UnquiesceSystem	✓ Successful

- **Net effect:** because step 4 aborts before Package/Save, **no backup file is written** to the SFTP target. There is no usable file-based backup until this clears.
- **Not a target/credential/NSX problem.** The backup gets past the resource lock (cleared separately) and past SFTP directory validation (/sddc on 192.168.1.160 validated OK). It is **not** related to the disconnected NSX accounts.

2. Root Cause (verified)

2.1 The crash

From the backup's own log (sos.log), the metadata exporter throws a Python `KeyError`:

```
ERROR [sddcmanagerbackup.py::export_metadata::287] 'primaryDatastoreName'
File "backup/sddcmanagerbackup.py", line 236, in export_metadata
File "backup/sddcmanagerbackup.py", line 237, in <lambda>
KeyError: 'primaryDatastoreName'
```

Immediately before the crash the exporter calls `GET http://localhost/v1/clusters?domainId=<mgmt>` (HTTP 200), then a lambda reads `cluster['primaryDatastoreName']` for each cluster — and the key is **absent** from the API response.

2.2 The database is already correct — the API is not

The platform DB row is fully populated, but the /v1/clusters API omits the fields. This is the crux: the fix is **not** a missing DB value.

Field	Platform DB (<code>cluster</code> table)	<code>/v1/clusters</code> API
name	vcenter-cl01 ✓	absent ✗
primary_datastore_name / primaryDatastoreName	vcenter-cl01-ds-vsan01 ✓	absent ✗
primary_datastore_type / primaryDatastoreType	VSAN	VSAN ✓

Field	Platform DB (<code>cluster</code> table)	<code>/v1/clusters</code> API
<code>datacenter</code>	—	<code>{}</code> empty ❌

The inventory API resolves `name` and `primaryDatastoreName` **through the cluster** → **datacenter** → **datastore relationship**, not directly from the columns. That relationship is **broken** (the API returns an **empty** `datacenter` object for the cluster), so those two fields cannot be resolved and are omitted — which is exactly what makes the backup's `export_metadata` lambda throw.

Origin: the vCenter/vSAN **redeploy** updated vCenter but left SDDC Manager's cluster inventory relationship out of sync. This matches **KB 429861** verbatim ("a previous datastore change was made directly in vCenter but was not reflected in the SDDC Manager database"), whose published resolution is **contact Broadcom Support**.

3. What Was Attempted (and the result)

Recorded so the same ground isn't re-covered. All read-only steps were non-destructive; the one write was guarded and reversible.

Attempt	Type	Result
Retry <i>Backup Now</i> (multiple)	UI	❌ Same <code>primaryDatastoreName</code> failure every time
Reconnect the 2 ESXi accounts (esxi01/04) first	Fix	✅ Reconnected, but did not change the backup outcome (not the cause)
Read <code>sos.log</code> for the real exception	Read-only	✅ Found <code>KeyError: 'primaryDatastoreName' at export_metadata</code>
GET <code>/v1/clusters/<id></code> to inspect the object	Read-only	✅ Confirmed API omits <code>name</code> + <code>primaryDatastoreName</code> ; <code>datacenter = {}</code>
Read the DB <code>cluster</code> row (trust-auth psql)	Read-only	✅ DB already correct (<code>primary_datastore_name = vcenter-cl01-ds-vsan01</code> , <code>name = vcenter-cl01</code>)
<code>information_schema</code> column discovery	Read-only	✅ Column is <code>cluster.primary_datastore_name</code> ; also <code>cluster.vsan_datastore_name</code> (was empty)
restart <code>operationsmanager</code> (twice)	Service	❌ API still omits the fields → not a stale cache
UPDATE <code>cluster</code> SET <code>vsan_datastore_name = 'vcenter-cl01-ds-vsan01'</code> (guarded)	DB write	❌ Populated the empty VSAN column; API still omits <code>primaryDatastoreName</code> → confirms it's the <i>relationship</i> , not the columns
Re-test backup after the above	UI	❌ Identical <code>KeyError: 'primaryDatastoreName'</code>

Conclusion of the attempts: every column the API *could* source the field from is now correct in the DB, and a service restart did not refresh the output — so the gap is the **cluster→datacenter→datastore relationship**, which cannot be repaired by setting a column. Rebuilding inventory relationships by hand would be unverified guesswork and was **deliberately not attempted**.

Footprint note. One reversible DB write was made: `cluster.vsan_datastore_name` was set from empty to its **correct** value `vcenter-cl01-ds-vsan01` (the real vSAN datastore name on the hosts). It is harmless and accurate to leave; to revert, set it back to empty (`' '`).

4. Verified Facts for Handoff

Hand these to Broadcom Support (or use them for a later reconciliation) so nothing has to be rediscovered.

Item	Value
Failing sub-task	<code>BackupSDDCManagerSystemConfiguration</code>
Error code	<code>BACKUP_SDDC_MGR_SYSTEM_CONFIG_FAILED</code> → <code>KeyError: 'primaryDatastoreName'</code>
Code location	<code>backup/sddcmanagerbackup.py</code> → <code>export_metadata</code> (≈ lines 236–237)
Cluster id	<code>dc5f4a5b-4ab2-41d9-9e7a-5e7d84ab2712</code> (<code>managedObjectReferenceId = domain-c10</code>)
Cluster name (DB)	<code>vcenter-cl01</code>
vSAN datastore (verified on host)	<code>vcenter-cl01-ds-<code>vsan01</code></code> (<code>vsan:52b68472c149d706-5c7eef39e7136ffa</code>)
DB columns (already set)	<code>cluster.primary_datastore_name</code> , <code>cluster.primary_datastore_type=VSAN</code> , <code>cluster.vsan_datastore_name</code>
API gap	<code>/v1/clusters/<id></code> omits <code>name + primaryDatastoreName</code> ; <code>datacenter = {}</code>
Backup log	<code>/var/log/vmware/vcf/sddc-support/backup-<timestamp>/sos.log</code>
Matching KB	KB 429861 (resolution = contact Support)

5. Options / Next Steps

1. **Broadcom Support (prescribed).** KB 429861 routes here; they repair the cluster inventory/datacenter relationship server-side. Requires a support entitlement.
2. **Inventory reconciliation.** The desync may clear as a side-effect of a future inventory resync (e.g., the planned **NSX Manager redeploy** + vCenter inventory refresh, or a domain/cluster re-import). Re-test the backup afterward before pursuing Support.
3. **Interim posture.** The **Database** and **Configuration** sub-backups *do* run; a manual/scripted copy of those is a partial stopgap, but it is **not** a substitute for a complete file-based backup. Treat the environment as **without a validated file-based backup** until this clears.

Do not hand-edit inventory relationship tables (datacenter/datastore associations) to force the field. There is no Broadcom-published procedure for it; an unverified edit risks corrupting inventory consistency. Keep this

on the supported path.

6. Document Information

Field	Value
Title	SDDC Manager Backup — System Configuration Fails on primaryDatastoreName [INCOMPLETE]
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Related Documents	VCF9-Operations-Campaign-2026-06-06-to-2026-06-19 (§3.4); RUNBOOK-SDDC-Manager-Resource-Lock-DB-Cleanup-2026-06-09
Reference KBs	429861 (this issue); 411099 (related backup-System-Config failure)
Distribution	Lab notebook / Support handoff

Revision History

Version	Date	Notes
1.0	2026-06-09	Initial record. Captured the <code>BackupSDDCManagerSystemConfiguration</code> → <code>KeyError: 'primaryDatastoreName'</code> failure, the verified root cause (DB correct but <code>/v1/clusters</code> API omits <code>name / primaryDatastoreName</code> via an empty <code>datacenter</code> relationship), the full attempt log (read-only diagnosis + one reversible DB write + service restarts), verified handoff facts, and options. Marked INCOMPLETE / Support-tier per KB 429861.

Document End — *Status: INCOMPLETE. This issue is not resolved.*